DATA ENGINEERING PROJECT

From raw CSV to live dashboard

A retail sales pipeline on Amazon S3, Snowflake (Snowpark) and Streamlit, written in Python

Jerry Wyatt

Swipe →

The data, in four numbers

$52.7K

total revenue

100

transactions, Jan to Apr 2023

3 + 4

products and regions

$527

average order value

A small dataset: this project is about pipeline design.

The pipeline, end to end

1

Amazon S3

Raw landing zone for sales.csv, loaded with boto3

2

Snowflake external stage

Reads straight from S3, no extra copy step

3

Snowpark (Python)

Cleans, derives and aggregates inside Snowflake

4

Streamlit in Snowflake

Dashboard queries the curated tables in place

Design choices

1

Transformations in Snowpark

Python DataFrames that Snowflake runs as SQL

2

External stage over S3

Snowflake reads the landing zone directly

3

CLI with typed errors

One command; every failure is logged and exits non-zero

4

A pandas twin

Same metrics, testable with no cloud account

What broke, and what I learned

1

.env never loaded

Nothing called load_dotenv(), so credentials came back empty

2

Someone else's bucket

S3 names are global; my placeholder name was already taken

3

Dot vs hyphen

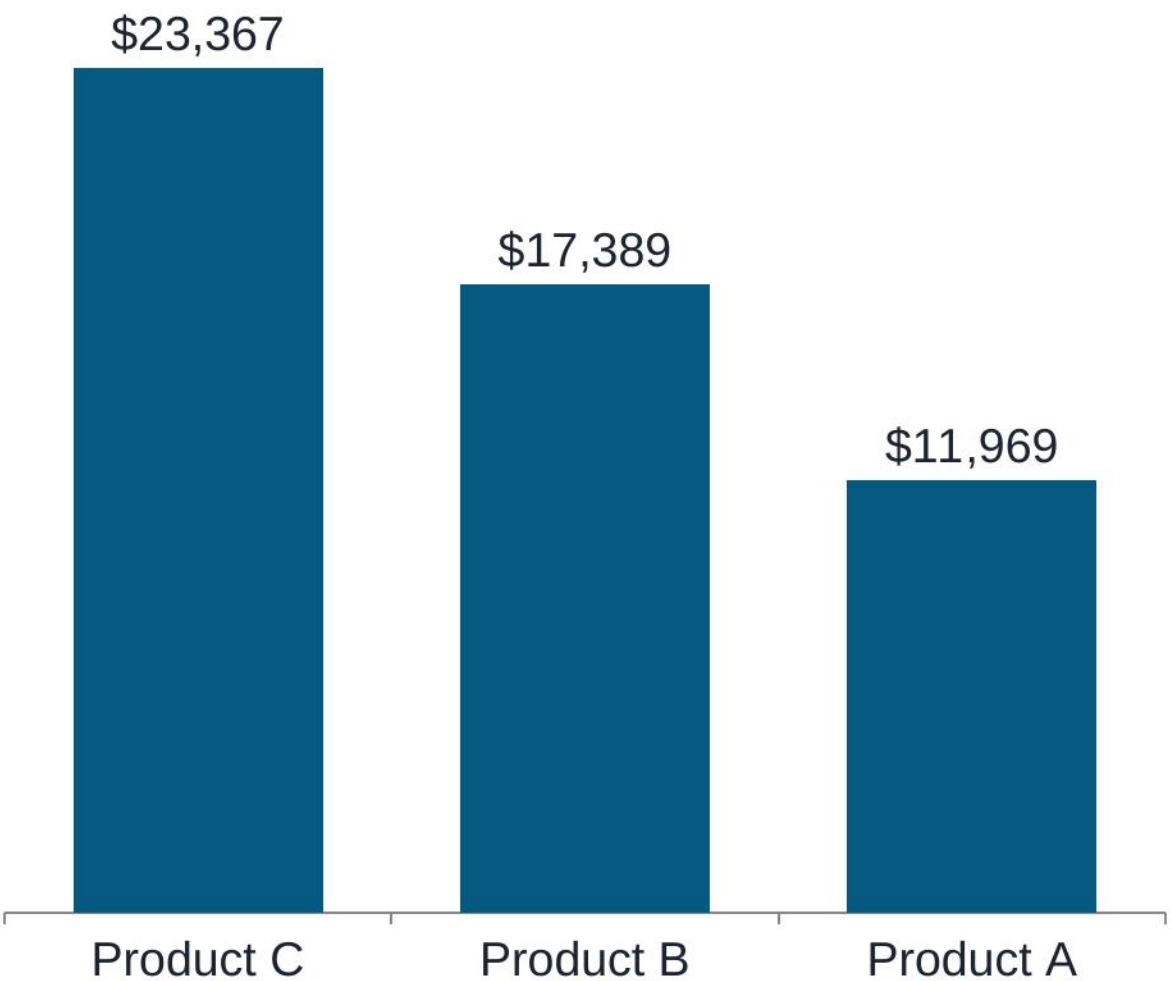
The account ID is org-account; a dot breaks the TLS hostname

4

A mislabeled metric

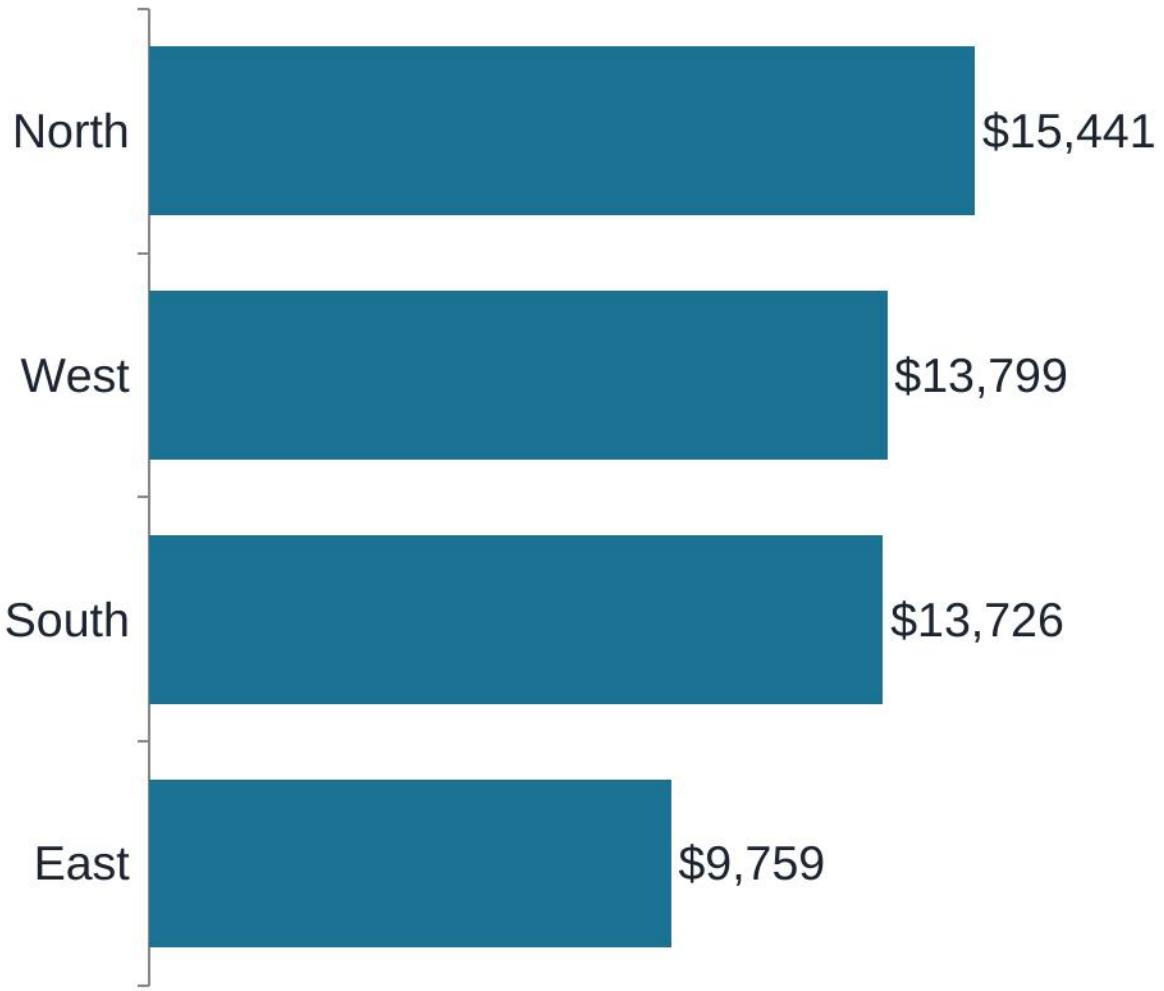
Spec's "AOV" was revenue per unit ($52.20). True AOV is $527.25

Product C drives 44% of revenue



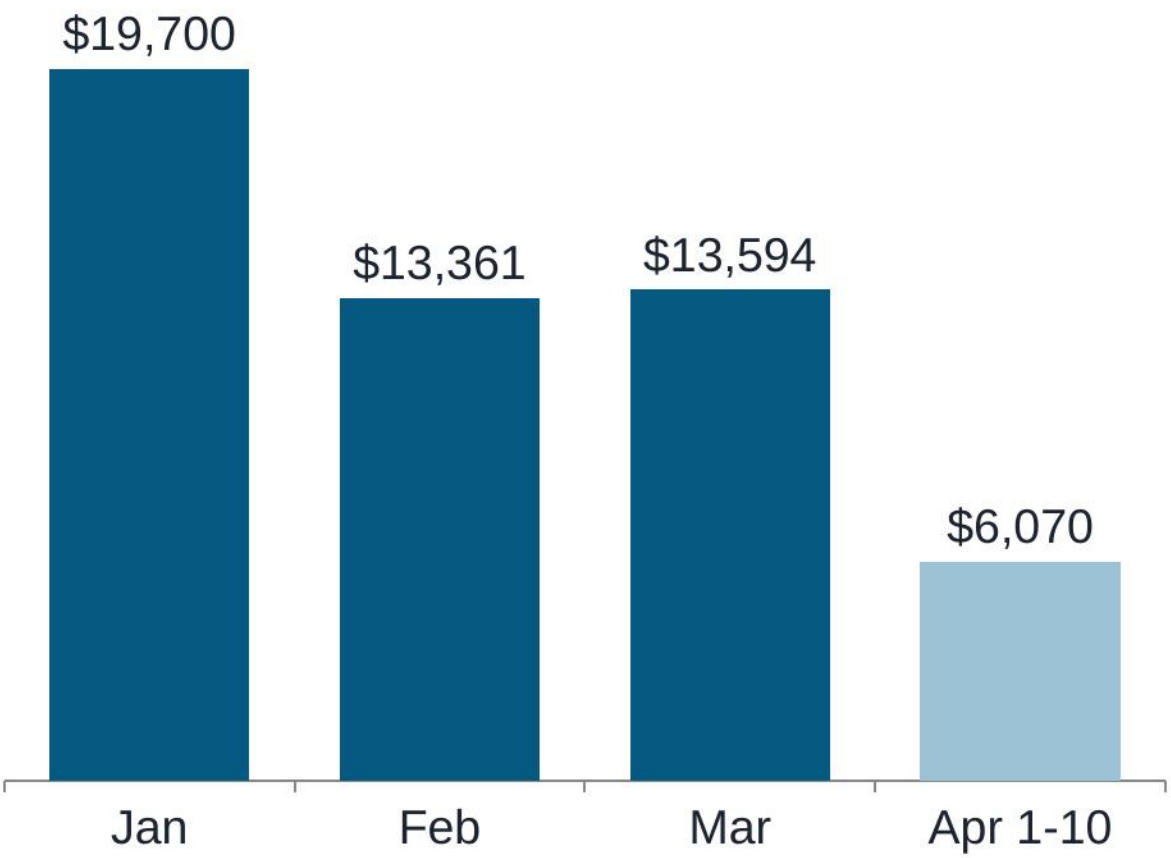
95% ahead of Product A, with the best revenue per unit ($54.98).

North leads, East lags



East trails North by 37%. Its average order value ($407) is the lowest too.

January peaked; April is partial



April has only 10 days of data, so that drop isn't a real decline.

STACK

Python · Snowpark · Snowflake
Amazon S3 · Streamlit · pandas

NEXT UP

- Storage integration, not embedded keys
- Scheduled runs with Snowflake tasks
- Unit tests on the metric logic

Thanks for reading.
Feedback welcome.